Basic Mechanics

-ME 162-

Moments

Department of Mechanical Engineering, KNUST

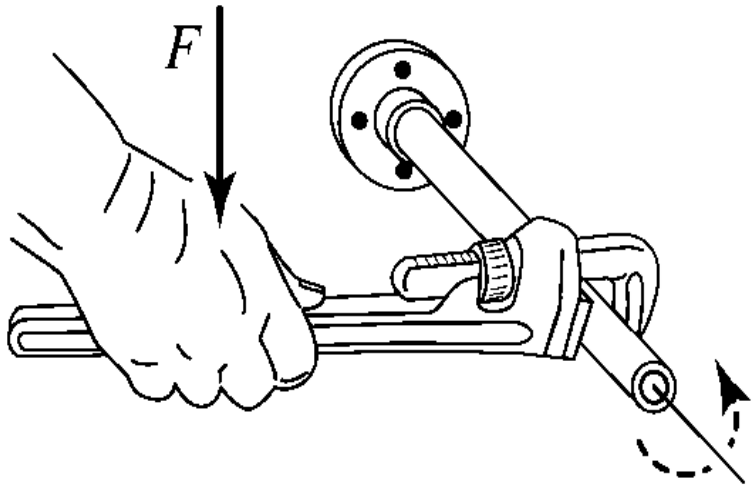
D. A. Quansah, PhD

Moments

- The ***moment of a force***, or simply ***moment***, is a measure of a force's ability to produce twisting, or rotation about a point.

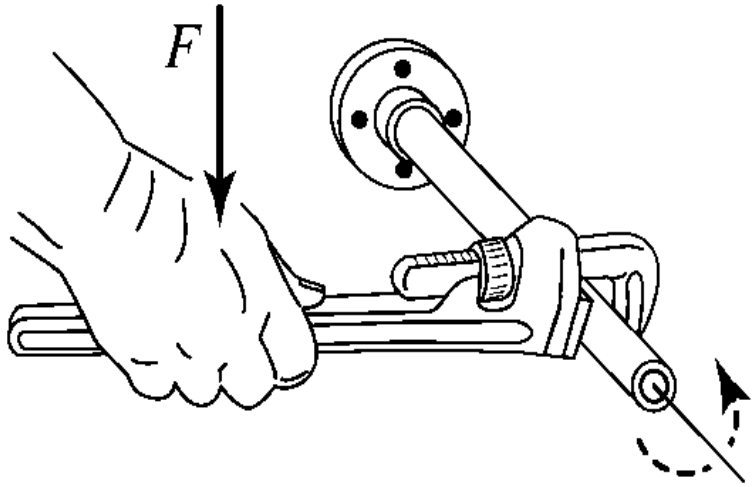
- To create a greater tendency to twist the pipe, either a larger force is applied to the wrench or a wrench with a longer handle is used.

- Moment is a **vector quantity** and has both **magnitude** and **direction**.

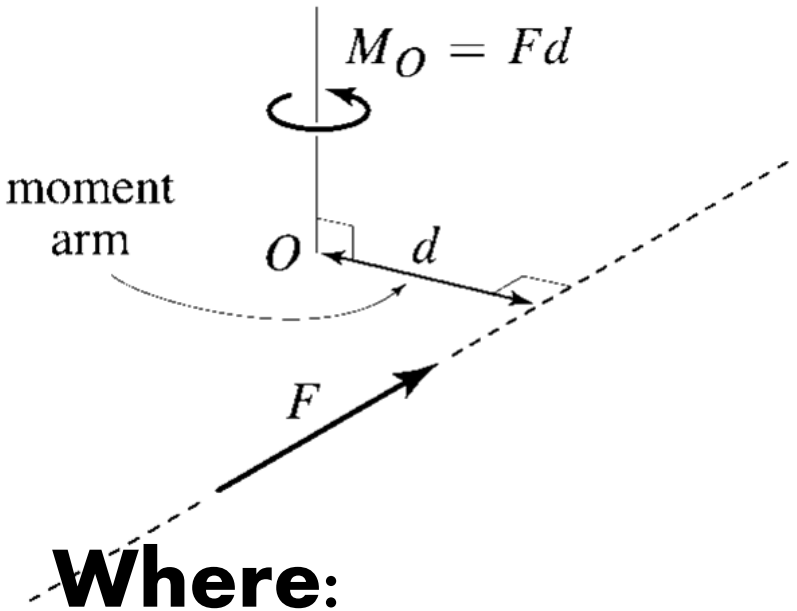


Moments

- **It can be evaluated using both scalar and vector approaches.**



Moments – Scalar Approach



- Consider a force $\vec{F}$ with magnitude ***F*** (shown in the diagram).
- The force produces a moment vector about point ***O***.
- The **magnitude** of this moment is **Mo** and is given by:

$$M_o = Fd$$

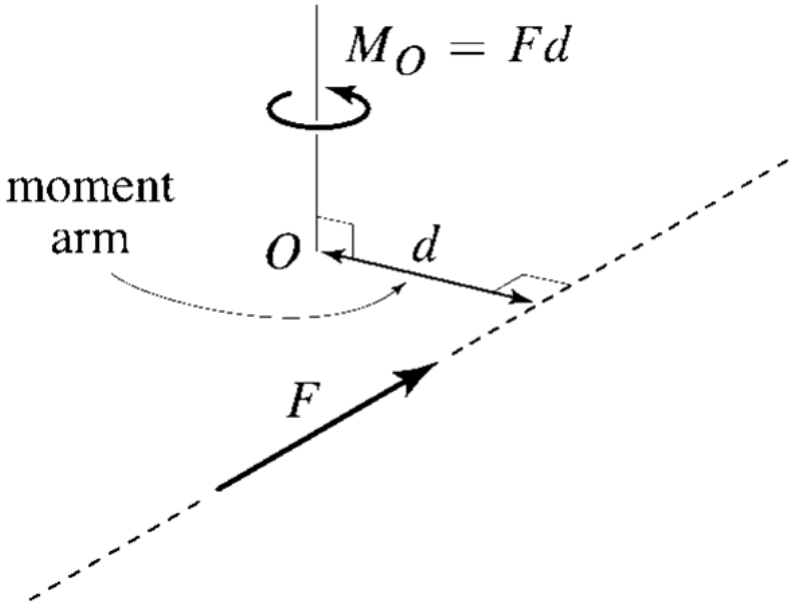
Where:

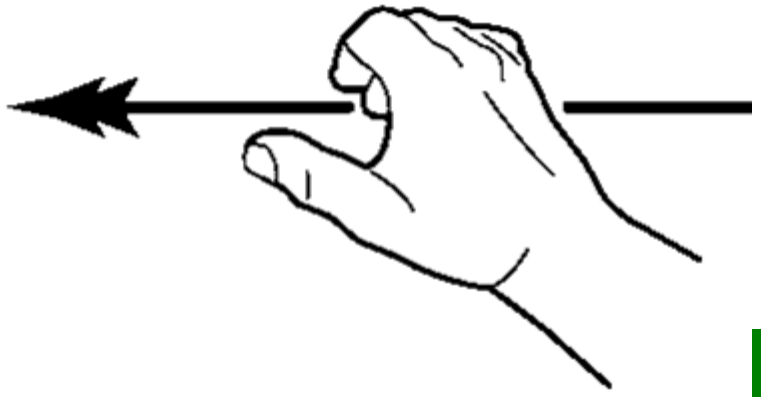
F is the magnitude of the force;

d is the perpendicular distance from point ***O*** to the line of action of $\vec{F}$.

d is called the **moment arm**.

Moments – Scalar Approach





- The **line of action** of the moment is parallel to the axis through point ***O*** that is **perpendicular to the plane containing $\vec{F}$ and the moment arm**.

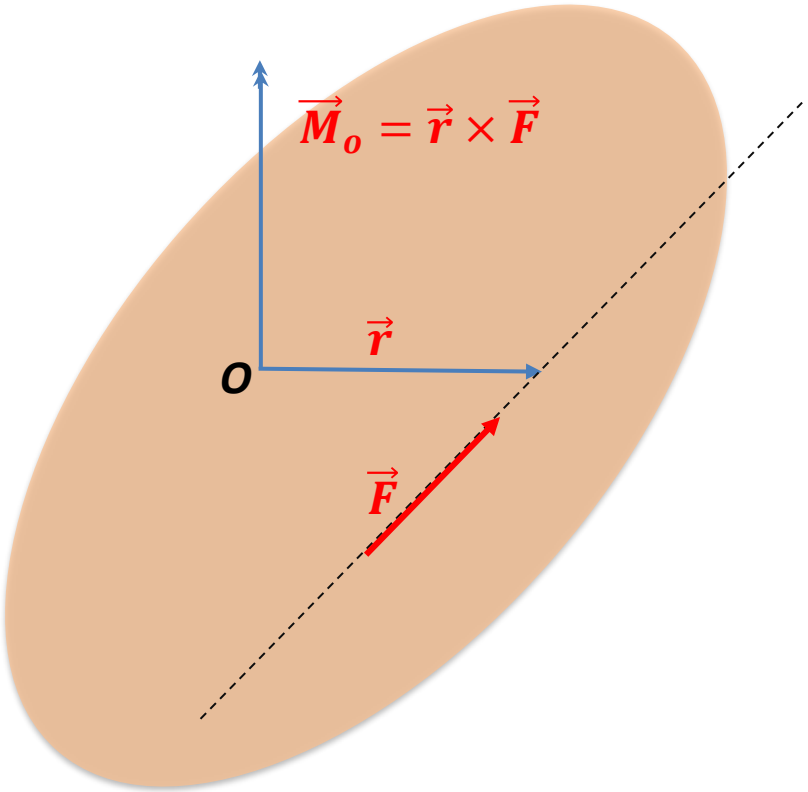
- The **direction of the moment** along the line of action is given by the direction of the thumb of your right hand when your fingers curl in the **twisting direction of the moment**.

Moments – Vector Approach

- The **magnitude** and **direction** of the moment of a force can be obtained using the **cross product**.
- The moment $\vec{M}_o$ of a force $\vec{F}$ about a point O is given by:

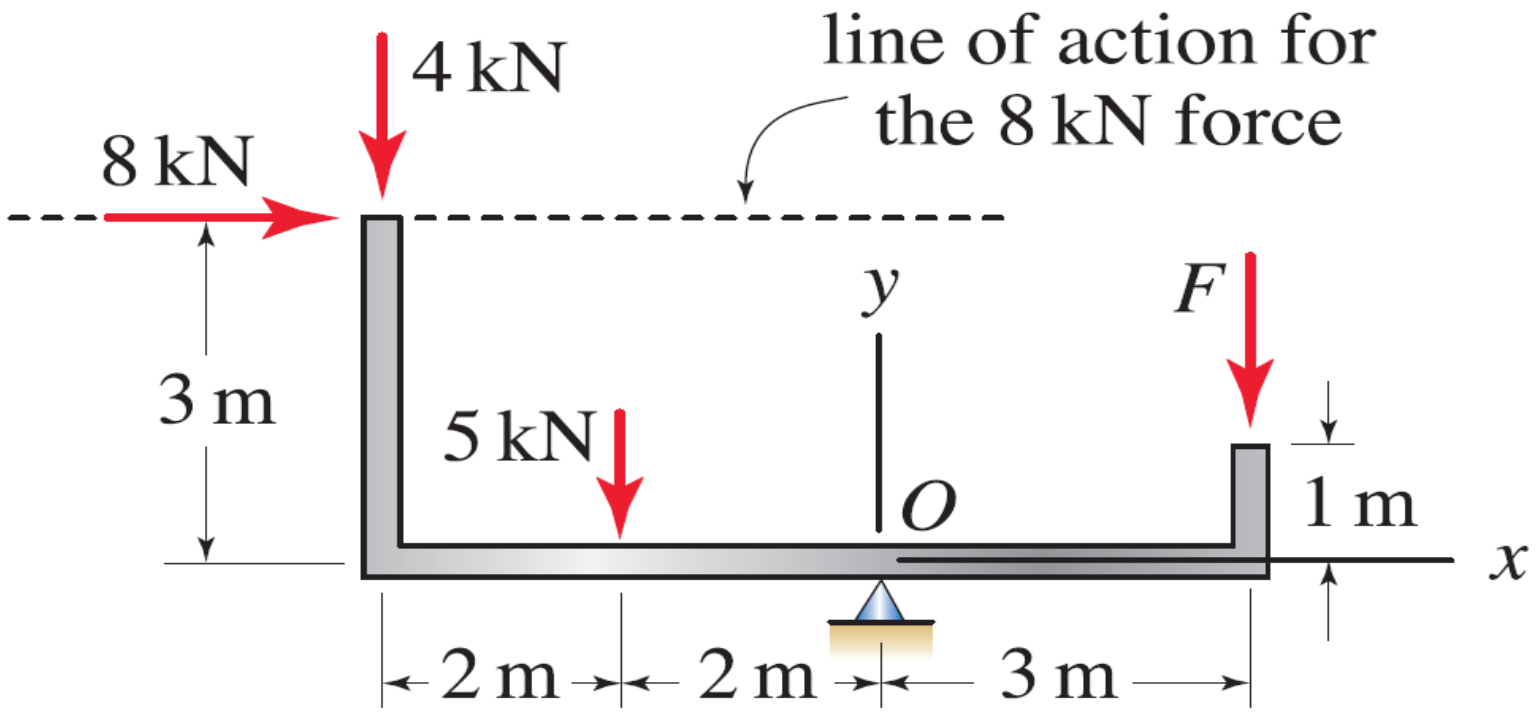
$$\vec{M}_o = \vec{r} \times \vec{F}$$

- Where $\vec{r}$ is a position vector from point ***O*** to **any point** on the line of action of the force $\vec{F}$.
- This approach automatically provides both the magnitude and direction of the moment.



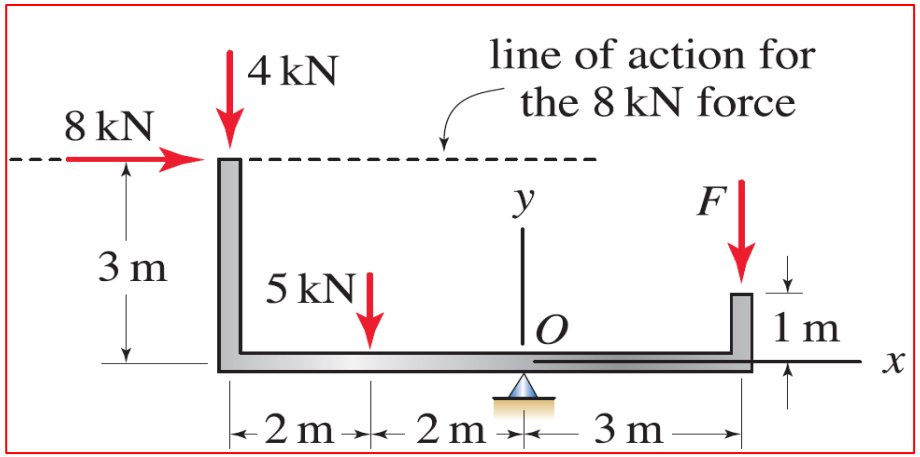
Worked Example

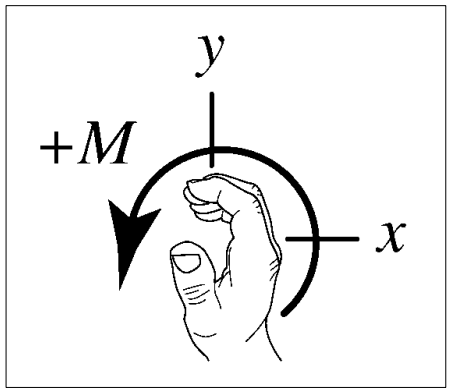
Example



For the balance mechanism shown, determine the value of *F* so that the resultant moment of all forces about point *O* is zero.

Example



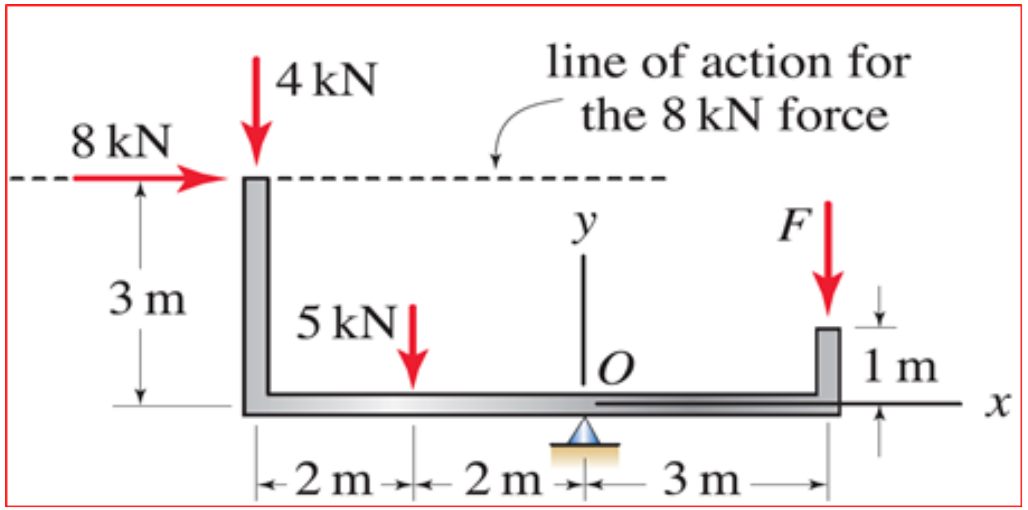


Scalar Approach:

- Note that each force has a tendency to twist the structure about point ***O*** in either a clockwise or counterclockwise fashion.
- The moment of each force about point ***O*** is a vector that points either into or out of the plane.

- Distinguish between these by taking the counterclockwise direction to be positive.

Example



$$\mathbf{M_o} = -(8\ \mathbf{kN})(3\ \mathbf{m}) + (4\ \mathbf{kN})(4\ \mathbf{m})$$

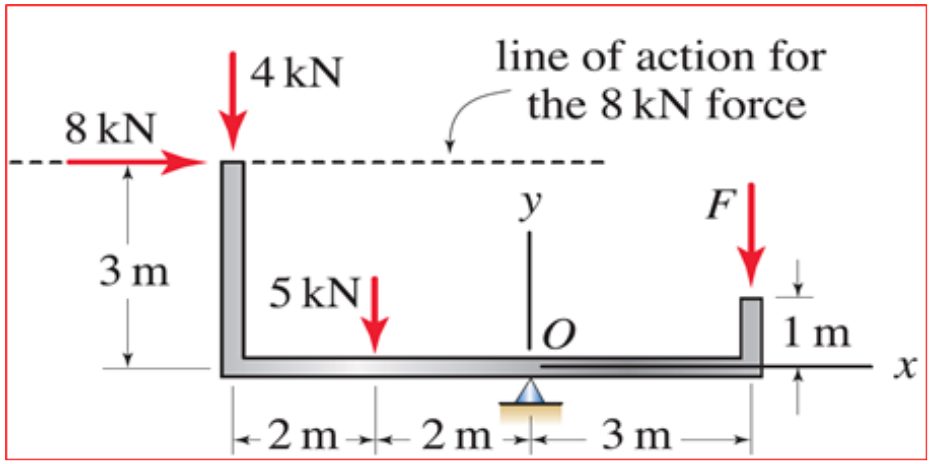
$$+(5\ \mathbf{kN})(2\ \mathbf{m}) - (\mathbf{F})(3\ \mathbf{m}) = 0$$

$$\mathbf{F} = 0.667\ \mathbf{kN}$$

COMMENTS:

- Each term consists of the product of the **magnitude of a force** with its corresponding **moment arm**.
- Where the **moment arm** is the shortest (perpendicular) distance from point ***O*** to the **line of action of the force**.

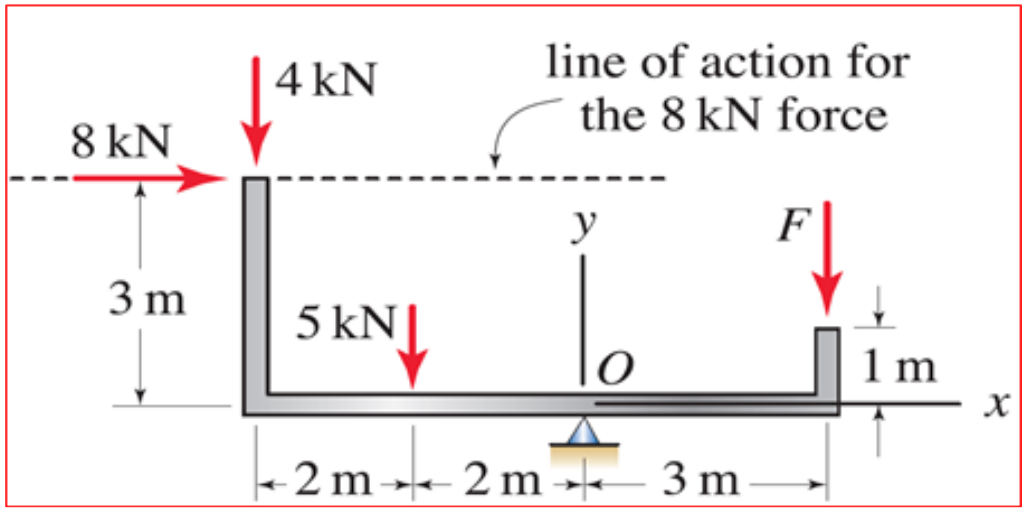
Example



COMMENTS (Ctd):

- **In the Figure above, the line of action for the 8 kN force is shown, and the shortest distance from point O to this line is 3 m.**

Example



$$M_o = -(8\ \text{kN})(3\ \text{m}) + (4\ \text{kN})(4\ \text{m})$$

$$+(5\ \text{kN})(2\ \text{m}) - (F)(3\ \text{m}) = 0$$

$$= 0.667\ \text{kN}$$

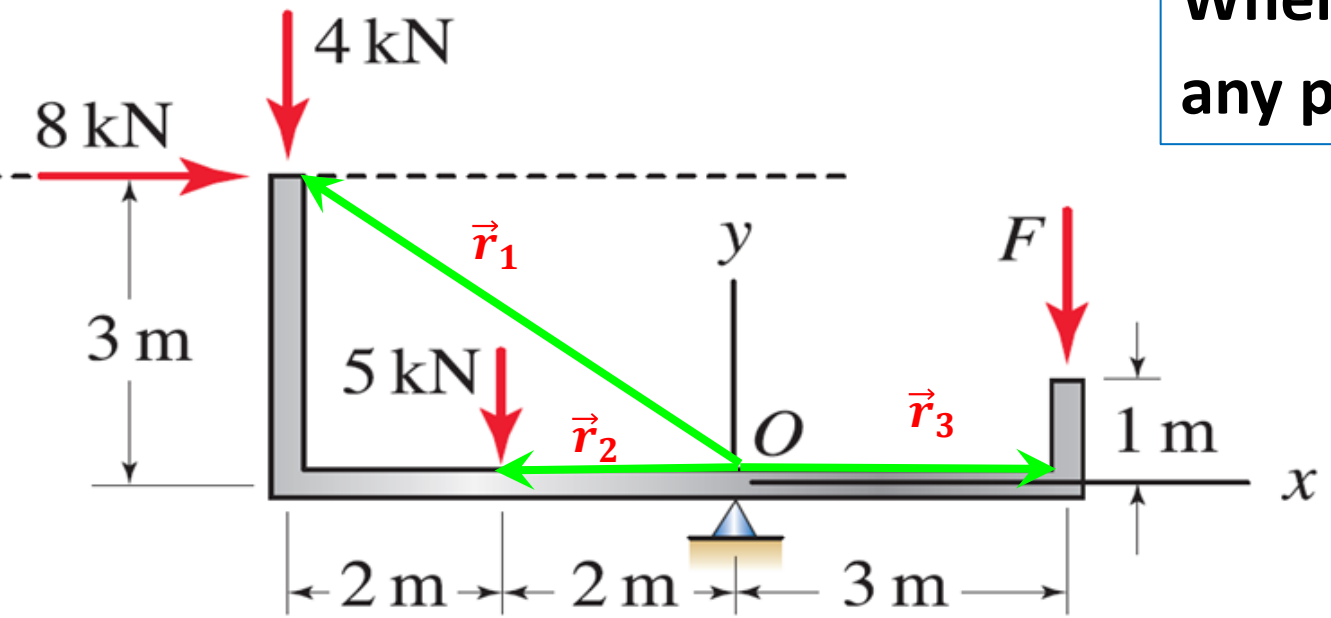
- The sign of each term is positive for counterclockwise twisting about point O and negative for clockwise twisting.

Vector Approach

Recall:

$$\vec{M}_o = \vec{r} \times \vec{F}$$

Where $\vec{r}$ is a position vector from point *O* to any point on the line of action of the force $\vec{F}$.



Force Vector	Position Vector
$\vec{F}_1 = (8i - 4j)\ kN$	$\vec{r}_1 = (-4i + 3j)\ m$
$\vec{F}_2 = -5j\ kN$	$\vec{r}_2 = -2i\ m$
$\vec{F}_3 = -Fj\ kN$	$\vec{r}_3 = 3i\ m$

$$\vec{M}_o = \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + \vec{r}_3 \times \vec{F}_3$$

$$\vec{M}_o = (-4i + 3j) \times (8i - 4j) + (-2i) \times (-5j) + (3i) \times (-Fj)$$

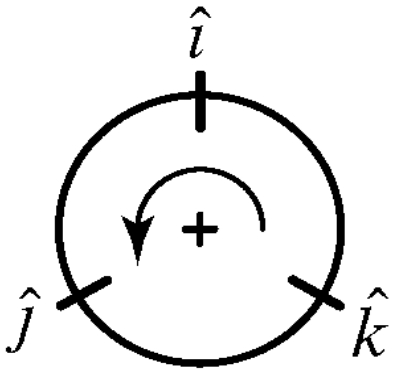
Vector Cross Products - Recall

$$\vec{A} = A_x\boldsymbol{i} + A_y\boldsymbol{j} + A_z\boldsymbol{k}$$

$$\vec{B} = B_x\boldsymbol{i} + B_y\boldsymbol{j} + B_z\boldsymbol{k}$$

$$\vec{A} \times \vec{B} = \left(\left|\vec{A}\right|\left|\vec{B}\right| \sin\theta\right)\hat{u}$$

Where $\hat{u}$ is a unit vector normal to the plane containing $\vec{A}$ and $\vec{B}$.



Cross Product of Unit Vectors:

$$\hat{\imath} \times \hat{\imath} = \left(\left|\vec{\hat{\imath}}\right|\left|\vec{\hat{\imath}}\right| \sin 0^o\right)\hat{u} = \hat{0}$$

$$\hat{\imath} \times \hat{\jmath} = \left(\left|\vec{\hat{\imath}}\right|\left|\vec{\hat{\jmath}}\right| \sin 90^o\right)\hat{u} = \hat{k}$$

$$\hat{\imath} \times \hat{\imath} = \hat{0} \qquad \hat{\jmath} \times \hat{\imath} = -\hat{k} \qquad \hat{k} \times \hat{\imath} = \hat{\jmath}$$

$$\hat{\imath} \times \hat{\jmath} = \hat{k} \qquad \hat{\jmath} \times \hat{\jmath} = \hat{0} \qquad \hat{k} \times \hat{\jmath} = -\hat{\imath}$$

$$\hat{\imath} \times \hat{k} = -\hat{\jmath} \qquad \hat{\jmath} \times \hat{k} = \hat{\imath} \qquad \hat{k} \times \hat{k} = \hat{0}$$

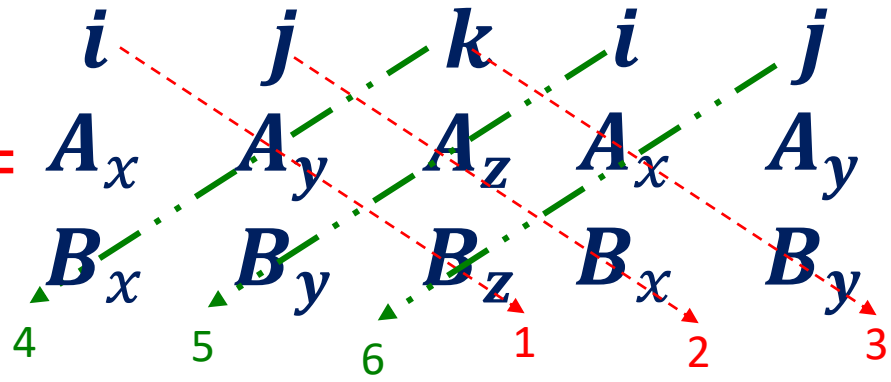
Vector Cross Products - Recall

$$\vec{A} = A_x i + A_y j + A_z k$$

$$\vec{B} = B_x i + B_y j + B_z k$$

$$\vec{A} \times \vec{B} =$$

$$= \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$



$$det \begin{bmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{bmatrix}$$

$$= \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= iA_yB_z + jA_zB_x + kA_xB_y$$
$$-kA_yB_x - iA_zB_y - jA_xB_z$$

$$= (A_yB_z - A_zB_y)i + (A_zB_x - A_xB_z)j$$
$$+(A_xB_y - A_yB_x)k$$

Vector Cross Products - Recall

$$\vec{A} \times \vec{B} = det \begin{bmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{bmatrix}$$

$$= i \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} - j \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} + k \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

$$= i(A_y B_z - A_z B_y) - j(A_x B_z - A_z B_x) + k(A_x B_y - A_y B_x)$$

Vector Cross Products - Recall

$$\overrightarrow{M_1} = \vec{r}_1 \times \vec{F}_1 = det\begin{bmatrix} i & j & k \\ r_{1x} & r_{1y} & r_{1z} \\ F_{1x} & F_{1y} & F_{1z} \end{bmatrix}$$

$$\vec{r}_1 \times \overrightarrow{F_1} = det\begin{bmatrix} i & j & k \\ -4 & 3 & 0 \\ 8 & -4 & 0 \end{bmatrix}$$

$$\begin{vmatrix} i & j & k \\ -4 & 3 & 0 \\ 8 & -4 & 0 \end{vmatrix} = \begin{matrix} i & j & k & i & j \\ -4 & 3 & 0 & -4 & 3 \\ 8 & -4 & 0 & 8 & -4 \end{matrix}$$

$$= i(3)(0) + j(0)(8) + k(-4)(-4) - k(3)(8) - i(0)(-4) - j(-4)(0)$$

$$\overrightarrow{M_1} = 16k - 24k = (-8\,k\,)kN.m$$

$$\overrightarrow{M_2} = \vec{r}_2 \times \vec{F}_2 = \begin{vmatrix} i & j & k \\ -2 & 0 & 0 \\ 0 & -5 & 0 \end{vmatrix}$$

$$\begin{matrix} i & j & k & i & j \\ -2 & 0 & 0 & -2 & 0 \\ 0 & -5 & 0 & 0 & -5 \end{matrix}$$

$$= i(0)(0) + j(0)(0) + k(-2)(-5) - k(0)(0) - i(0)(-5) - j(-2)(0)$$

$$= (10k)\,kN.m$$

$$\overrightarrow{M_3} = \vec{r}_3 \times \vec{F}_3 = \begin{vmatrix} i & j & k \\ 3 & 0 & 0 \\ 0 & -F & 0 \end{vmatrix}$$

$$\begin{matrix} i & j & k & i & j \\ 3 & 0 & 0 & 3 & 0 \\ 0 & -F & 0 & 0 & -F \end{matrix}$$

Vector Cross Products - Recall

$$\vec{M}_3 = \vec{r}_3 \times \vec{F}_3 = \begin{vmatrix} i & j & k \\ 3 & 0 & 0 \\ 0 & -F & 0 \end{vmatrix}$$

$$\begin{matrix} i & j & k & i & j \\ 3 & 0 & 0 & 3 & 0 \\ 0 & -F & 0 & 0 & -F \end{matrix}$$

$$= i(0)(0) + j(0)(0) + k(3)(-F) - k(0)(0) - i(0)(-F) - j(3)(0)$$

$$= (-3Fk)\ kNm$$

$$\vec{M}_o = \vec{M}_1 + \vec{M}_2 + \vec{M}_3 = \vec{0}$$

$$= (-8\,k)kN.m + (10k)kN.m + (-3\,k)\mathrm{F}.m$$

$$[2\ \mathrm{kN.m} - (3\ \mathrm{m})\mathrm{F}]k = \vec{0}$$

$$\mathrm{F} = 0.667\ \mathrm{kN}$$

Varignon's Theorem (Principle of Moments)

- The moment of a force is equal to the **sum of the moments of the vector components of the force**.
- Therefore if a force $\vec{F}$ has components $\vec{F}_1$, $\vec{F}_2$, etc., then the moment of F about a point A is given as:

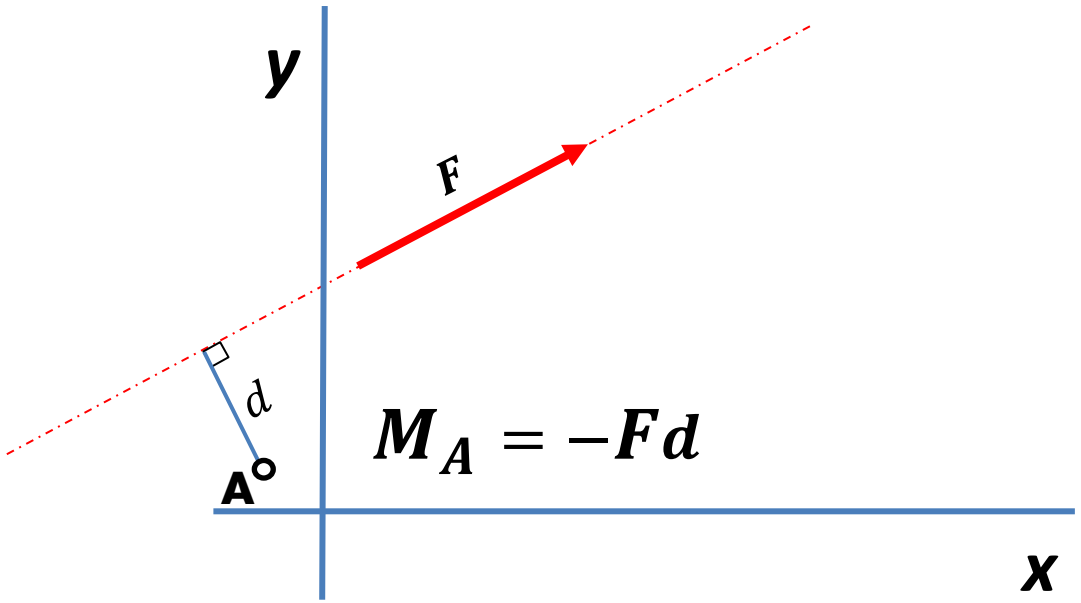
$$\vec{M} = \vec{r} \times \vec{F}$$

$$\vec{M} = \vec{r} \times (\vec{F}_1 + \vec{F}_2 + \cdots)$$

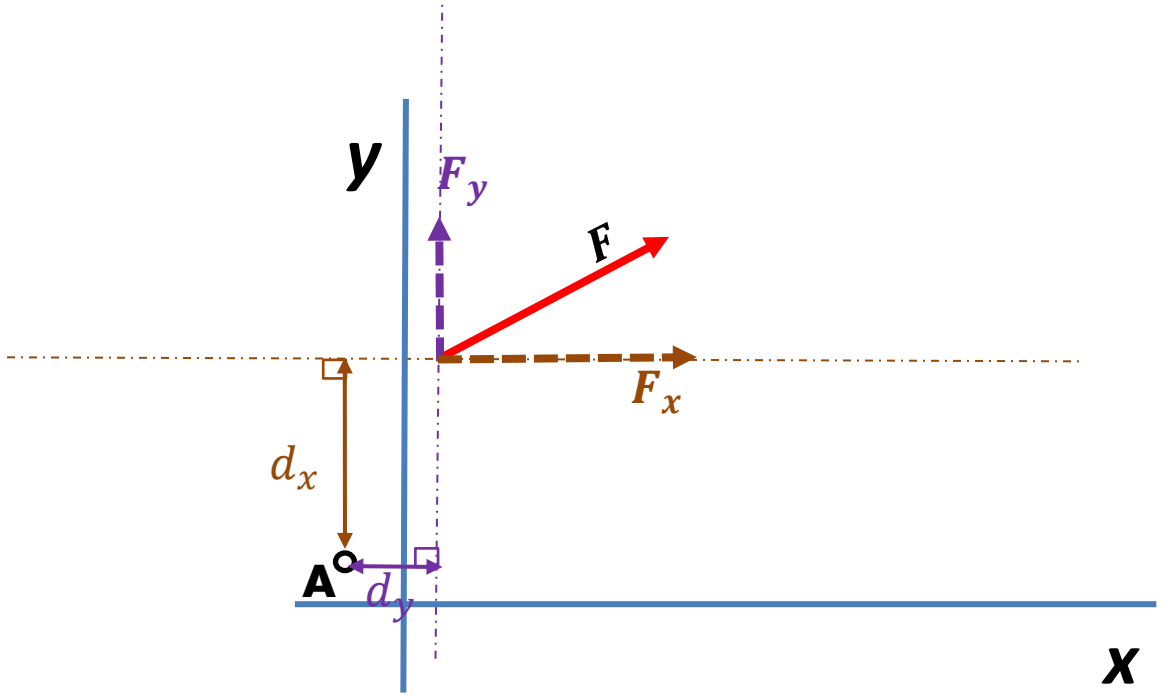
$$\vec{M} = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2 + \cdots$$

- Where $\vec{r}$ is a position vector from point A to any point on the line of action of $\vec{F}$.

Scalar Description of Varignon's Theorem



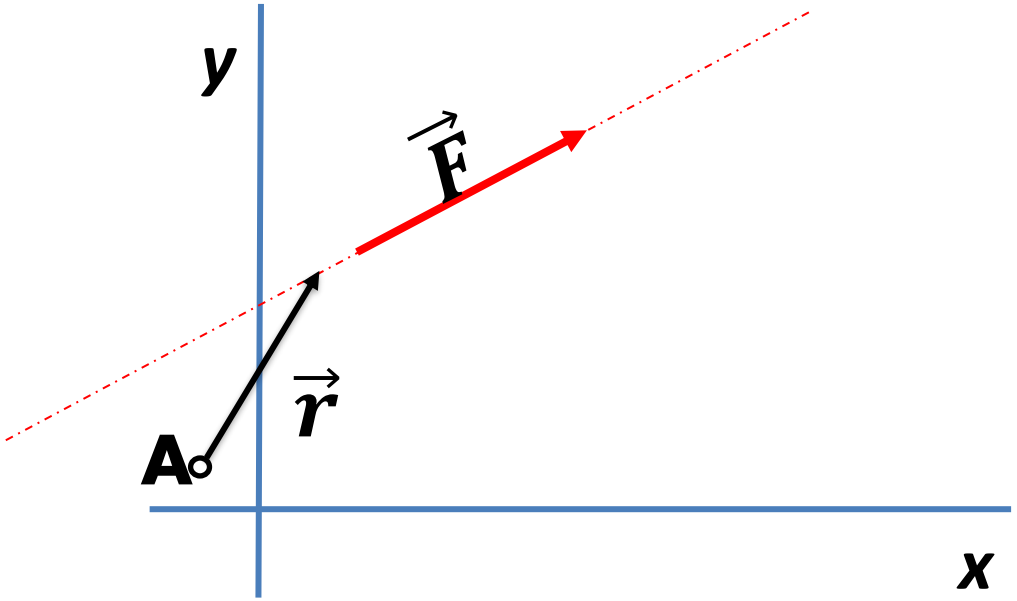
$$M_A = -Fd$$



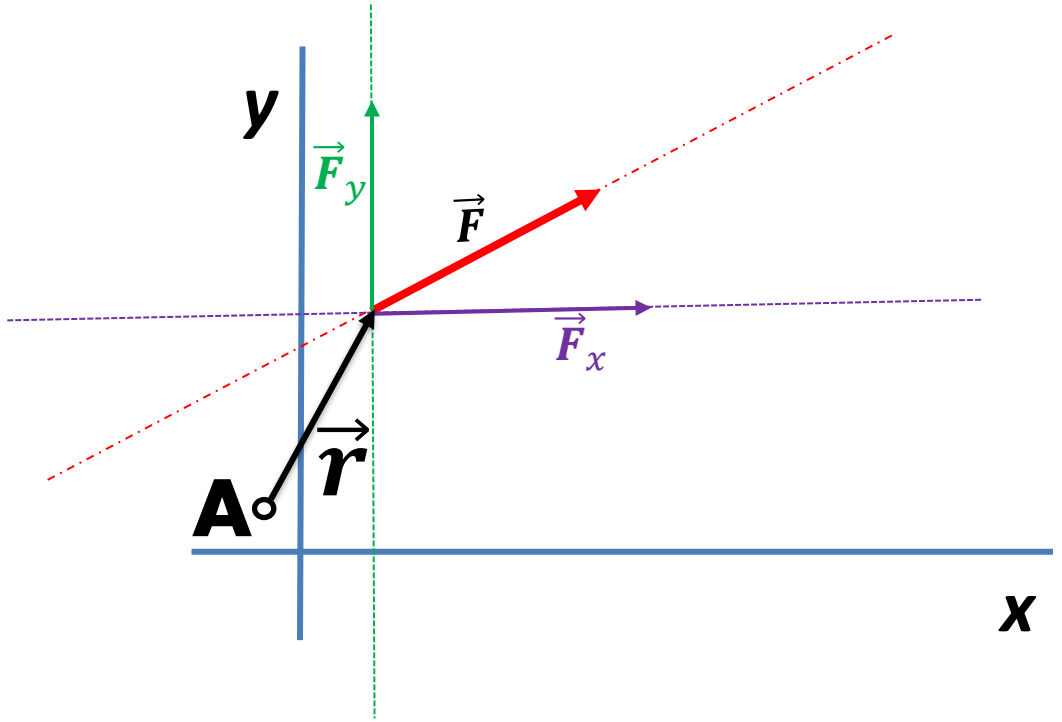
Sign convention:
clockwise is –ve, anticlockwise is +ve

$$M_A = -F_x d_x + F_y d_y$$

Vector description of Varignon's theorem

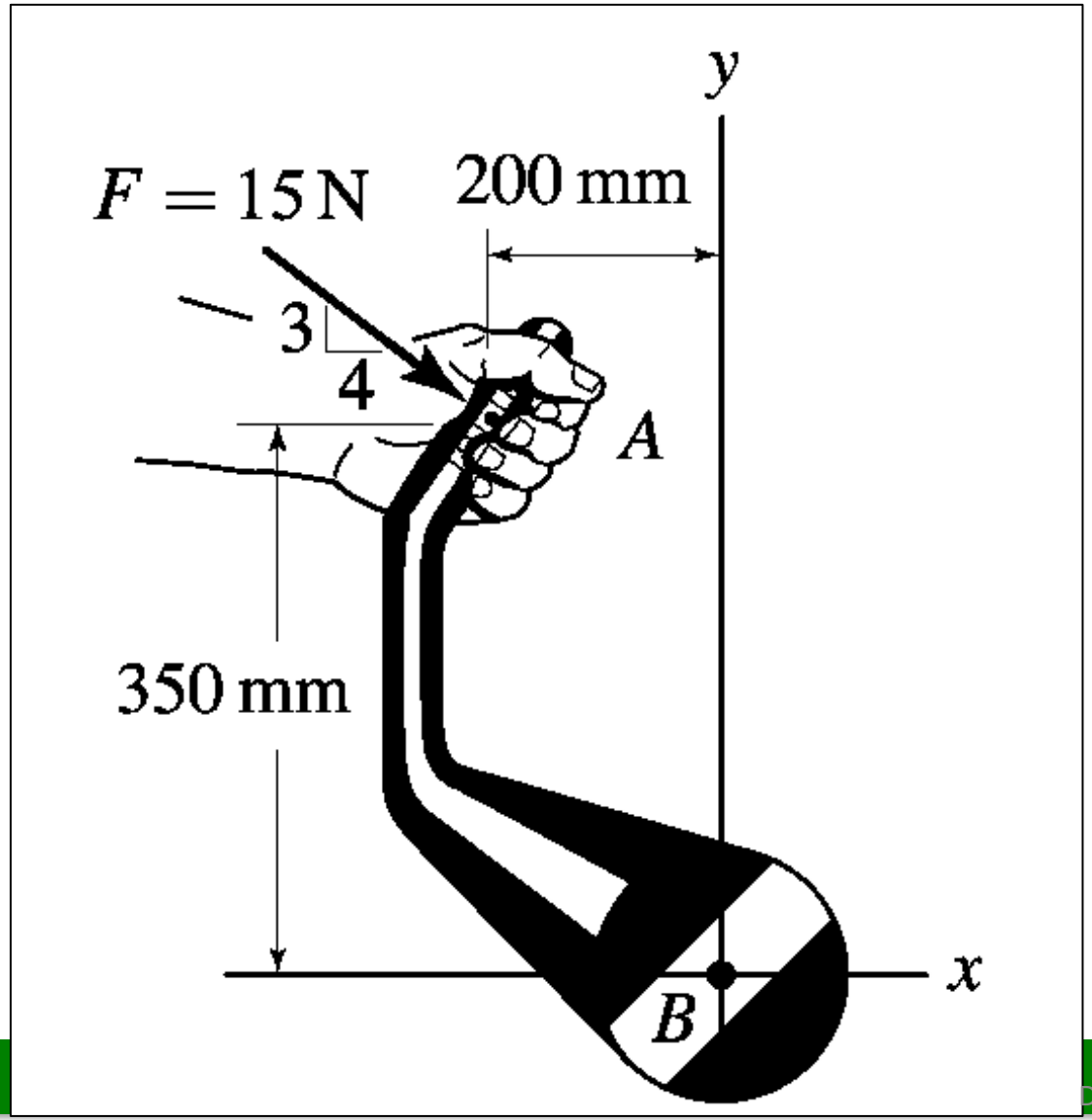


$$\vec{M}_A = \vec{r} \times \vec{F}$$



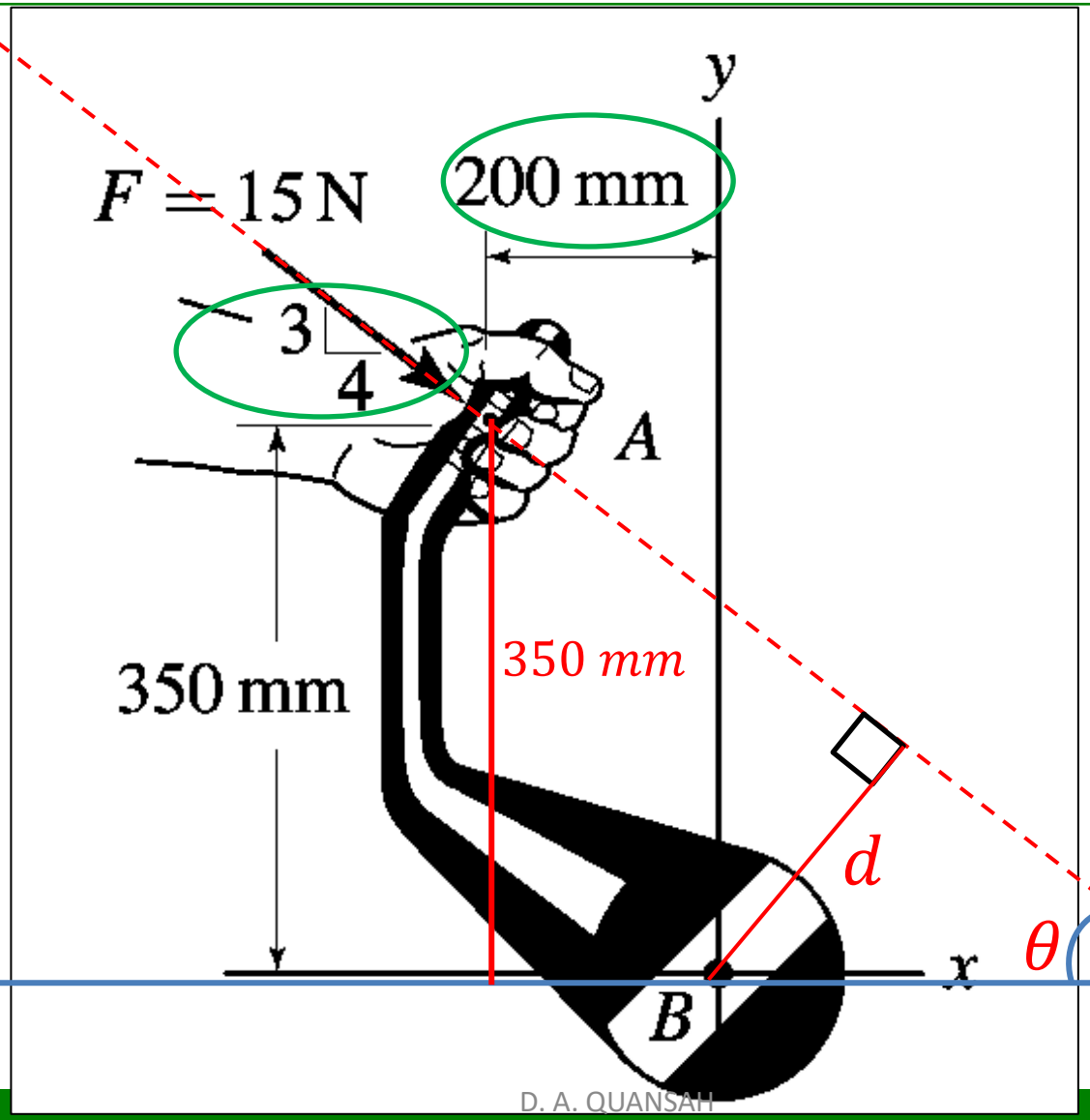
$$\vec{M}_A = \vec{r} \times \vec{F}_x + \vec{r} \times \vec{F}_y$$

EXAMPLE



A machine handle is connected to a shaft at B. Determine the moment produced by the 15 N force about point B.

EXAMPLE



Please try your hands on this!